

#1 What is the total number of customers, total conversions, and overall conversion rate (in percentage)?

```
select
count(customer_id) as Total_Customers,
sum(case when conversion='Yes' then 1 else 0 end) as Total_Conversion,
(sum(case when conversion='Yes' then 1 else 0 end)*100)/count( customer_id)
as Conversion_Rate
from table1;
```

	Total_Customers	Total_Conversion	Conversion_Rate
▶	8000	7012	87.6500

#2 What is the average advertising spend, average CTR, average conversion rate, and average website engagement across all customers?

```
select
avg(ad_spend) as Average_Advertising_Spend,
avg(click_through_rate) as Average_CTR,
avg(conversion_rate) as Average_Conversion_Rate,
avg(time_on_site) as Average_Website_Engagement
from table1;
```

	Average_Advertising_Spend	Average_CTR	Average_Conversion_Rate	Average_Website_Engagement
▶	5000.944830424923	0.1548286491509563	0.10438873502114518	7.727718162515271

#3 What percentage of customers converted versus did not convert?

```
select
(sum(case when conversion='Yes' then 1 else 0 end)*100)/count(customer_id) as
Converted_Customers_Percentage,
(sum(case when conversion='No' then 1 else 0 end)*100)/count(customer_id) as
Non_Converted_Customers_Percentage
from table1;
```

	Converted_Customers_Percentage	Non_Converted_Customers_Percentage
▶	87.6500	12.3500

#4 What is the conversion rate (in percentage) for each Gender?

```
select gender,  
(sum(case when conversion='Yes' then 1 else 0 end)*100)/count(customer_id) as  
Conversion_Rate  
from table1  
group by gender;
```

	gender	Conversion_Rate
▶	Female	87.6214
	Male	87.6938

#5 Which age_group has the highest conversion rate (in percentage) and how many customers does it contain?

```
select  
age_group as Age_Group,  
count(customer_id) as Total_Customers,  
(sum(case when conversion='Yes' then 1 else 0 end)*100)/count(customer_id)  
as Conversion_Rate  
from table1  
group by age_group  
order by Conversion_Rate desc;
```

	Age_Group	Total_Customers	Conversion_Rate
▶	Adult	1939	87.9319
	Senior	1999	87.6938
	Young Adult	2078	87.6323
	Middle Aged	1984	87.3488

#6 How does conversion rate (in percentage) vary across income_group?

```
select  
income_group as Income_Group,  
(sum(case when conversion='Yes' then 1 else 0 end)*100)/count(customer_id)  
as Conversion_Rate  
from table1  
group by Income_Group;
```

	Income_Group	Conversion_Rate
►	High	88.1515
	Low	87.2891
	Medium	87.5094

#7 How do average income, average ad spend, and average conversion rate differ between converted and non-converted customers?

```
select
avg(income) as Average_Income,
avg(ad_spend) as Average_Ad_Spend,
avg(conversion_rate) as Average_Conversion_Rate,
conversion as Conversion_group
from table1
group by conversion;
```

	Average_Income	Average_Ad_Spend	Average_Conversion_Rate	Conversion_group
►	84861.3019	5133.750849839879	0.10630818157760247	Yes
	83265.3087	4058.398465913321	0.09076610419738107	No

#8 Which campaign_channel generates the highest number of conversions and the highest conversion rate (in percentage)?

```
select campaign_channel,
sum(case when conversion='Yes' then 1 else 0 end) as Conversions,
(sum(case when conversion='Yes' then 1 else 0 end)*100)/count(customer_id) as
Conversion_Rate
from table1
group by campaign_channel
order by Conversions desc;
```

	campaign_channel	Conversions	Conversion_Rate
►	Referral	1518	88.3072
	PPC	1461	88.2779
	SEO	1359	87.6774
	Email	1355	87.0263
	Social Media	1319	86.8334

#9 Which campaign_type generates the highest number of conversions and the highest conversion rate (in percentage)?

```

select campaign_type,
sum(case when conversion='Yes' then 1 else 0 end)/count(customer_id)
as Number_Of_Conversions,
(sum(case when conversion='Yes' then 1 else 0 end)*100)/count(customer_id)
as Conversion_Rate
from table1
group by campaign_type
order by Number_Of_Conversions desc;

```

	campaign_type	Number_Of_Conversions	Conversion_Rate
►	Conversion	0.9336	93.3558
	Retention	0.8582	85.8243
	Awareness	0.8556	85.5634
	Consideration	0.8556	85.5634

#10 What is the performance of each campaign_channel in terms of customer count,

total ad spend, conversions, CTR, and conversion rate(in percentage)?

```

select campaign_channel,
count(customer_id) as Number_Of_Customers,
sum(ad_spend) as Total_Ad_Spend,
sum(case when conversion='Yes' then 1 else 0 end)/count(customer_id)
as Number_Of_Conversions,
avg(click_through_rate) as Average_CTR,
(sum(case when conversion='Yes' then 1 else 0 end)*100)/count(customer_id)
as Conversion_Rate
from table1
group by campaign_channel;

```

	campaign_channel	Number_Of_Customers	Total_Ad_Spend	Number_Of_Conversions	Average_CTR	Conversion_Rate
►	Social Media	1519	7542323.249154089	0.8683	0.15552270740622332	86.8334
	Email	1557	7871575.851879107	0.8703	0.15560905231967895	87.0263
	PPC	1655	8199236.981122514	0.8828	0.15830726306304424	88.2779
	Referral	1719	8653518.685591804	0.8831	0.15167256449745575	88.3072
	SEO	1550	7740903.875652119	0.8768	0.1531504822292855	87.6774

#11 What is the performance of each campaign_type in terms of customer count,

total ad spend, conversions, CTR, and conversion rate (in percentage) ?

```

select campaign_type,
count(customer_id) as Number_Of_Customers,
sum(ad_spend) as Total_Ad_Spend,
sum(case when conversion='Yes' then 1 else 0 end)/count(customer_id)
as Number_Of_Conversions,
avg(click_through_rate) as Average_CTR,
(sum(case when conversion='Yes' then 1 else 0 end)*100)/count(customer_id)
as Conversion_Rate
from table1
group by campaign_type;

```

	campaign_type	Number_Of_Customers	Total_Ad_Spend	Number_Of_Conversions	Average_CTR	Conversion_Rate
►	Awareness	1988	10077845.826437416	0.8556	0.15576413973963363	85.5634
	Retention	1947	9768362.161081562	0.8582	0.15607586320072972	85.8243
	Conversion	2077	10300076.981282627	0.9336	0.1556650848936359	93.3558
	Consideration	1988	9861273.674598029	0.8556	0.1517977849242234	85.5634

#12 Which combination of campaign_channel and campaign_type has the highest conversion rate (in percentage)?

```

select
campaign_channel,
campaign_type,
(sum(case when conversion='Yes' then 1 else 0 end)*100)/count(customer_id)
as Conversion_Rate
from table1
group by
campaign_channel,
campaign_type
order by Conversion_Rate desc;

```

	campaign_channel	campaign_type	Conversion_Rate
►	SEO	Conversion	94.0299
	PPC	Conversion	93.9597
	Email	Conversion	93.2692
	Referral	Conversion	93.0337
	Social Media	Conversion	92.3706
	Referral	Awareness	88.3991
	PPC	Consideration	86.8812
	SEO	Retention	86.6324
	Social Media	Retention	86.2162
	Referral	Retention	86.0092
	PPC	Awareness	85.9223
	PPC	Retention	85.7143
	Email	Consideration	85.6079
	Referral	Consideration	85.5037
	Social Media	Consideration	84.9383
	SEO	Awareness	84.8718
	SEO	Consideration	84.8238
	Email	Retention	84.4444
	Email	Awareness	84.1270
	Social Media	Awareness	84.0849

#13 Which combination of campaign_channel and campaign_type generates the highest number of conversions?

```

select
campaign_channel,
campaign_type,
sum(case when conversion='Yes' then 1 else 0 end) as Number_Of_Conversions
from table1
group by
campaign_channel,
campaign_type
order by Number_Of_Conversions desc;

```

	campaign_channel	campaign_type	Number_Of_Conversions
►	PPC	Conversion	420
	Referral	Conversion	414
	Email	Conversion	388
	Referral	Awareness	381
	SEO	Conversion	378
	Referral	Retention	375
	PPC	Awareness	354
	PPC	Consideration	351
	Referral	Consideration	348
	Email	Consideration	345
	Social Media	Consideration	344
	Social Media	Conversion	339
	SEO	Retention	337
	PPC	Retention	336
	SEO	Awareness	331
	Social Media	Retention	319
	Email	Awareness	318
	Social Media	Awareness	317
	SEO	Consideration	313
	Email	Retention	304

#14 What is the cost per conversion for each campaign_channel?

```
select campaign_channel,
sum(ad_spend)/sum(case when conversion='Yes' then 1 else 0 end)
as Cost_Per_Conversion
from table1
group by campaign_channel;
```

	campaign_channel	Cost_Per_Conversion
►	Social Media	5718.213229078157
	Email	5809.281071497496
	PPC	5612.071855662228
	Referral	5700.605194724509
	SEO	5696.029341907372

#15 What is the cost per conversion for each campaign_type?

```
select campaign_type,
sum(ad_spend)/sum(case when conversion='Yes' then 1 else 0 end)
as Cost_Per_Conversion
from table1
group by campaign_type;
```

	campaign_type	Cost_Per_Conversion
►	Awareness	5924.659509957329
	Retention	5845.818169408475
	Conversion	5312.056204890473
	Consideration	5797.339020927707

#16 Which campaign channel has above-average ad spend but below-average conversion rate?

with ChannelPerformance as

```
(select
campaign_channel,
sum(ad_spend) as Total_Ad_Spend,
avg(conversion_rate) as Average_conversion_Rate
from table1
group by campaign_channel),
DatasetBenchmark as
(select
avg(Total_Ad_Spend) as Benchmark_Spend,
avg(Average_Conversion_Rate) as Benchmark_Conversion_Rate
from ChannelPerformance)
select
cp.campaign_channel,
cp.Total_Ad_Spend,
cp.Average_conversion_Rate
from ChannelPerformance cp
cross join DatasetBenchmark db
where
cp.Total_Ad_Spend>db.Benchmark_Spend
and
cp.Average_Conversion_Rate<db.Benchmark_Conversion_Rate;
```

	campaign_channel	Total_Ad_Spend	Average_conversion_Rate
►	PPC	8199236.981122514	0.10413227300416425
	Referral	8653518.685591804	0.10305104397231174

#17 Which campaign channel has below-average ad spend but above-average conversion rate?

with ChannelPerformance as

```

(select campaign_channel,
sum(ad_spend) as Total_Ad_Spend,
avg(conversion_rate) as Avg_Conversion_Rate
from table1
group by campaign_channel),
DatasetBenchmark as
(select avg(Total_Ad_Spend) as Benchmark_Spend,
avg(Avg_Conversion_Rate) as Benchmark_Avg_Conversion_Rate
from ChannelPerformance)
select
cp.campaign_channel,
cp.Total_Ad_Spend,
cp.Avg_Conversion_Rate
from
ChannelPerformance as cp
cross join
DatasetBenchmark as db
where
cp.Total_Ad_Spend < db.Benchmark_Spend
and
cp.Avg_Conversion_Rate > db.Benchmark_Avg_Conversion_Rate;

```

	campaign_channel	Total_Ad_Spend	Avg_Conversion_Rate
►	Social Media	7542323.249154089	0.10655013521066009
	Email	7871575.851879107	0.1048840253741055

#18 How does conversion rate change across different levels of advertising spend?

```

select case
when ad_spend=0 then '0 USD'
when ad_spend between 1 and 1000 then '1-1000 USD'
when ad_spend between 1001 and 2000 then '1001-2000 USD'
when ad_spend between 2001 and 3000 then '2001-3000 USD'
when ad_spend between 3001 and 4000 then '3001-4000 USD'
when ad_spend between 4001 and 5000 then '4001-5000 USD'
when ad_spend between 5001 and 6000 then '5001-6000 USD'
when ad_spend between 6001 and 7000 then '6001-7000 USD'
when ad_spend between 7001 and 8000 then '7001-8000 USD'
when ad_spend>8000 then '8000+ USD'

```

```

end as ad_spend_tier,
avg(conversion_rate) as Avg_Conversion_Rate
from table1
group by ad_spend_tier
order by ad_spend_tier asc;

```

	ad_spend_tier	Avg_Conversion_Rate
▶	HULL	0.08282388256053444
	1-1000 USD	0.10562126424469498
	1001-2000 USD	0.10406968540706811
	2001-3000 USD	0.10316063130513513
	3001-4000 USD	0.10956928638788105
	4001-5000 USD	0.10713046779792669
	5001-6000 USD	0.10364404641550469
	6001-7000 USD	0.10507362600107671
	7001-8000 USD	0.10132552418382315
	8000+ USD	0.10210102098272221

#19 How does website engagement differ between converted and non-converted customers?

#Consider website_visits, pages_per_visit, and time_on_site.

```

select
conversion,
avg(website_visits) as avg_website_visits,
avg(pages_per_visit) as avg_page_per_visit,
avg(time_on_site) as avg_time_on_site
from table1
group by
conversion;

```

	conversion	avg_website_visits	avg_page_per_visit	avg_time_on_site
▶	Yes	25.1778	5.649944693237983	7.933412504876044
	No	21.7267	4.8350015679089555	6.2678712711856175

#20 How does email engagement differ between converted and non-converted customers?

Consider email_opens and email_clicks.

```

select
conversion,
avg(email_opens) as avg_email_opens,
avg(email_clicks) as avg_email_clicks

```

from table1

group by conversion;

	conversion	avg_email_opens	avg_email_clicks
►	Yes	9.7446	4.6062
	No	7.5769	3.4818

#21 How does previous purchase behaviour affect conversion rate?

select

case

when previous_purchase=0 then '0 previous purchase'

when previous_purchase between 1 and 3 then '1-3 previous purchases'

when previous_purchase between 4 and 10 then '4-10 previous purchases'

else '10+ previous purchases'

end as purchase_tier,

count(*) as Total_Customers,

avg(conversion_rate) as Avg_Conversion_Rate

from table1

group by purchase_tier;

	purchase_tier	Total_Customers	Avg_Conversion_Rate
►	4-10 previous purchases	4773	0.1033557232309659
	1-3 previous purchases	2389	0.10504401103028223
	0 previous purchase	838	0.10840438047305528

#22 How does loyalty-point level affect conversion rate?

select

count(customer_id) as Total_Customers,

case

when loyalty_points between 1 and 500 then '1-500 points (Bronze)'

when loyalty_points between 501 and 1500 then '501-1500 points (Silver)'

when loyalty_points between 1501 and 3000 then '1501-3000 points (Gold)'

else '3000+ points (Diamond)'

end as loyalty_tier,

avg(conversion_rate) as Avg_Conversion_Rate

from table1

group by loyalty_tier;

	Total_Customers	loyalty_tier	Avg_Conversion_Rate
►	1638	501-1500 points (Silver)	0.10601369317878631
	3166	3000+ points (Diamond)	0.10446308495229716
	2433	1501-3000 points (Gold)	0.10359777311414874
	763	1-500 points (Bronze)	0.10311394730880061

#23 Do customers with higher website engagement, email engagement, and previous purchases have higher conversion rates?

```
select
conversion,
avg(click_through_rate) as avg_ctr,
avg(time_on_site) as avg_time_spent,
avg(website_visits) as avg_website_visit,
avg(email_opens) as avg_email_opens,
avg(email_clicks) as avg_email_click,
avg(previous_purchase) as avg_previous_purchase
from table1
group by conversion;
```

	conversion	avg_ctr	avg_time_spent	avg_website_visit	avg_email_opens	avg_email_click	avg_previous_purchase
►	Yes	0.15861283628325937	7.933412504876044	25.1778	9.7446	4.6062	4.6067
	No	0.12797164492857777	6.2678712711856175	21.7267	7.5769	3.4818	3.6255

#24 Rank all campaign_channel values by conversion rate and show their total conversions and total ad spend.

```
select
campaign_channel,
avg(conversion_rate),
sum(ad_spend) as Total_Ad_Spend,
sum(case when conversion='Yes' then 1 else 0 end) as Total_Conversions
from table1
group by campaign_channel;
```

	campaign_channel	avg(conversion_rate)	Total_Ad_Spend	Total_Conversions
►	Social Media	0.10655013521066009	7542323.249154089	1319
	Email	0.1048840253741055	7871575.851879107	1355
	PPC	0.10413227300416425	8199236.981122514	1461
	Referral	0.10305104397231174	8653518.685591804	1518
	SEO	0.10353041346218893	7740903.875652119	1359

